

Matching Lines: Slope and Intercept

PhET Graphing Lines · PhET

Years 8-9

~30 minutes

Mathematics · Algebra & Functions

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/graphing-lines>

Curriculum links: AC9M8A02 · AC9M9A03 · linear equations

Learning intentions

- I can graph a linear equation given in slope-intercept form.
- I can find the gradient of a line from two points or from its graph.
- I can write the equation of a line when given its graph.

Getting started

Open PhET Graphing Lines and start on the 'Slope-Intercept' screen. Use the sliders to change the values of m (slope) and b (y-intercept) and watch how the line on the graph changes. Once you're comfortable, move to the 'Point-Slope' or 'Line Game' screens for the later tasks.

Tasks

1. Set the equation to $y = 2x + 1$. Predict where the line will cross the y-axis before you look, then check.

2. Keeping b the same, change m from 2 to -2. Describe what happens to the direction of the line.

3. Set b to 0 and try three different values of m . What do all three lines have in common?

4. Test four different combinations of slope and y-intercept and record what the line looks like for each.

Slope (m)	Y-intercept (b)	Steep or shallow?

5. Explain, in your own words, what the slope of a line tells you about how y changes as x increases. Use one of your examples from the table.

6. Switch to the 'Point-Slope' screen. Choose two points and read off the gradient the simulation calculates. Calculate the gradient yourself using rise over run and check it matches.

7. Two points on a line are (1, 4) and (5, 12). Calculate the gradient of the line between them by hand, then build this line in the simulation to check.

Working:

8. Use a matching or 'write the equation' challenge in the simulation to match a target line by adjusting only the equation values, not by trial and error. Describe the strategy you used to figure out the correct m and b.

Extension (optional)

Two lines are parallel if they have the same slope. Use the simulation to find two different lines that are parallel to each other, and explain how you knew they would never meet.
